

QF620 Stochastic Modelling in Finance — Session 2

Reference Cheatsheet

Binomial Tree and the Risk-Neutral Measure: Radon–Nikodym derivative, arithmetic and geometric trees, American options, conditional expectation, martingales, risk-neutral pricing

Built from Session 2 slides (Tee Chyng Wen, SMU) · September 2026

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Source: “Session 2: Binomial Tree and the Risk-Neutral Measure” (Tee Chyng Wen, SMU, 25 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. $\triangle!$ marks a slide ambiguity or error, with the corrected version.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From this session you need: the Radon–Nikodym derivative and how to reweight expectations; the no-arbitrage condition $d < 1+r < u$ and p^* ; backward induction for European and American options on a small tree; the two CLT limits (normal and lognormal trees); conditional-expectation rules; and why discounted prices are Q -martingales.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
Ω, ω	sample space; one outcome (a price path)	set; element	[S2]
$\mathbb{P}$	real-world (empirical) measure	probability	e.g. fund manager's 80% up [S4]
$\mathbb{Q}$	risk-neutral measure	probability	also written with $*$: $p^*, \mathbb{E}^*$
$\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega)$	Radon–Nikodym derivative	random variable	$= \mathbb{Q}(\omega)/\mathbb{P}(\omega)$ on a finite space [S2]
$S_n, S_{n+1}^u, S_{n+1}^d$	stock at step n ; up/down successors	price	
u, d	up and down factors	ratio	$u = S_1(H)/S_0, d = S_1(T)/S_0$ [S11]
r	risk-free rate per period (discrete)	rate	$\$1 \rightarrow \$(1+r)$ [S10]; $\triangle!$ continuous in [S15–S17]
p^*, q^*	risk-neutral up/down probabilities	probability	$p^* = \frac{(1+r)-d}{u-d}, q^* = 1 - p^*$ [S19]
σ	volatility	per $\sqrt{\text{time}}$	price units (arithmetic tree) or log units (geometric tree)
T, n or N	maturity; number of steps	—	step size T/n
X_i	i.i.d. ± 1 with probability $\frac{1}{2}$ each	random	drives the tree [S7, S15]
Z	standard normal	random	CLT limit [S8]
V_n, V_n^E, V_n^A	derivative value at step n ; European; American	price	[S13, S23]
$\mathbb{E}_n^*[\cdot]$	conditional expectation under $\mathbb{Q}$ given the first n tosses	random	[S19]
M_n	discounted price $S_n/(1+r)^n$	random	a $\mathbb{Q}$ -martingale [S21]
$(x)^+$	$\max(x, 0)$	—	payoff notation

Notation trap: [S10–S13, S19–S23] use a *discrete, per-period* rate $(1+r)$; [S15–S17] use a *continuously compounded* rate (e^{rT}) . The symbol r means different things in the two halves.

0. Topic map

PREREQUISITES

S1 Expectation, variance, MGF of the normal $E[e^{\theta X}] = e^{\{\mu\theta + \sigma^2\theta^2/2\}}$; lognormal $E[e^X] = e^{\{\mu + \sigma^2/2\}}$; two-state hedging: $\Delta = (C_u - C_d)/(S_u - S_d)$; risk-neutral p^* reproduces the hedge price

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SESSION 2: BINOMIAL TREE AND THE RISK-NEUTRAL MEASURE

└ RADON-NIKODYM [S2–S6]

- └ $(\Omega, \mathbb{P})$; $d\mathbb{P}/d\mathbb{Q}(\omega) = \mathbb{P}(\omega)/\mathbb{Q}(\omega)$ [S2]
- └ Needs equivalence: $\mathbb{P}(\omega) > 0 \Leftrightarrow \mathbb{Q}(\omega) > 0$ [S3]
- └ Example: $\mathbb{P}$ up 0.8, $\mathbb{Q}$ up 0.5, 100 $\pm$ 10 over two steps; $d\mathbb{Q}/d\mathbb{P}$ tree [S4]
- └ $E^*\mathbb{P}[S_2] = 112, E^*\mathbb{Q}[S_2] = 100; E^*\mathbb{Q}[X] = E^*\mathbb{P}[X \cdot d\mathbb{Q}/d\mathbb{P}]$ [S5–S6]

└ TREE MODELS [S7–S17]

- └ Arithmetic tree: $\pm\sigma\sqrt{T/n}$, $\text{prob } \frac{1}{2} \Rightarrow S_T = S_0 + \sigma\sqrt{T} Z$ (CLT) [S7–S9]
- └ $r \neq 0$: $S_d < S_0(1+r) < S_u$; $p^* = (S_0(1+r) - S_d)/(S_u - S_d)$ [S10]
- └ Geometric tree: u, d ; $p^* = (1+r-d)/(u-d)$; CRR $d = 1/u$ [S11]
- └ Example: European put, $S_0=4, u=2, d=\frac{1}{2}, r=25\%, K=5, 2$ steps [S12]
- └ American: $V = \max(\text{continuation}, \text{exercise})$ [S13]
- └ CLT for products $\Rightarrow$ lognormal [S14]
- └ Log tree: $\ln S_T = \ln S_0 + (r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} Z$; $E[S_T] = S_0 e^{rT}$ [S15–S17]

- └ P AND Q [S18–S20]
 - └ Coin example: random variables need no measure; distributions do [S18]
 - └ $E^*_n[S_{n+1}] = pS^u + qS^d$; $S_n = E^*_n[S_{n+1}]/(1+r)$ [S19]
 - └ Linearity, taking out what is known, tower rule, independence [S20]
- └ MARTINGALES & ARBITRAGE [S21–S25]
 - └ $M_n = S_n/(1+r)^n$ is a Q-martingale; $M_0 = E^*[M_N]$ [S21–S22]
 - └ Risk-neutral pricing formula $V_n = E^*[V_N/(1+r)^{N-n}]$ [S23]
 - └ Complete markets, replication, no-arbitrage $\Rightarrow$ Q exists; numeraires [S24–S25]

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LATER: S3 Brownian motion = limit of the $\pm\sqrt{t/n}$ random walk; continuous-time martingales;
 S4 Itô calculus; later Girsanov (continuous-time Radon–Nikodym) and Black–Scholes

Dependency logic:

- Session 1's one-period p^* becomes a whole probability measure $\mathbb{Q}$ on paths. The Radon–Nikodym derivative is the exchange rate between $\mathbb{P}$ and $\mathbb{Q}$ expectations.
- Backward induction with p^* at every node prices any payoff; American options add a max at each node.
- The CLT turns many small tree steps into a normal (arithmetic) or lognormal (geometric) terminal distribution: the bridge to Brownian motion (S3) and Black–Scholes.
- Conditional-expectation rules show that discounted prices are martingales under $\mathbb{Q}$, which is exactly the risk-neutral pricing formula.

1 Concept entries

1.1 Probability spaces and the Radon–Nikodym derivative [S2–S3]

- **Probability space** [S2]: $(\Omega, \mathbb{P})$. Ω lists what can happen; $\mathbb{P}$ decides how likely each outcome is. “ Ω lists the outcomes, $\mathbb{P}$ determines the distribution.”
- **Changing measure** [S2]: keeps the same paths but changes their likelihoods.
- **Radon–Nikodym derivative** [S2] (finite Ω):

$$\frac{d\mathbb{P}}{d\mathbb{Q}}(\omega) = \frac{\mathbb{P}(\omega)}{\mathbb{Q}(\omega)}, \quad \frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)}.$$

- **Equivalence** [S3]: $\mathbb{P}$ and $\mathbb{Q}$ are equivalent if they live on the same Ω and agree on what is possible: $\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0$. Only then are both derivatives defined everywhere.
- **Why it matters** [S3]: dropping outcomes that are $\mathbb{P}$ -impossible but $\mathbb{Q}$ -possible loses information about $\mathbb{Q}$. Equivalence is the formal version of Session 1's "only zero vs non-zero probabilities matter".
- **Properties** [Beyond slides]:
 - $\frac{d\mathbb{Q}}{d\mathbb{P}} > 0$ and $\mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}}\right] = \sum_{\omega} \mathbb{Q}(\omega) = 1$.
 - $\frac{d\mathbb{P}}{d\mathbb{Q}} = \left(\frac{d\mathbb{Q}}{d\mathbb{P}}\right)^{-1}$.
- **⚠** [S3] says that if $\mathbb{P}(\omega_i) = 0$ but $\mathbb{Q}(\omega_i) \neq 0$ then $\mathbb{P}(\omega_i)/\mathbb{Q}(\omega_i)$ is undefined. That ratio is 0 (defined); it is $d\mathbb{Q}/d\mathbb{P} = \mathbb{Q}/\mathbb{P}$ that breaks. The conclusion (you need equivalence for both directions) is right.
- **⚠** [S2] A probability space is properly a triple $(\Omega, \mathcal{F}, \mathbb{P})$ with a set of events $\mathcal{F}$; the slides drop $\mathcal{F}$ because Ω is finite.

1.2 Radon–Nikodym example: bullish fund manager vs market maker [S4–S6]

- **Setup [S4]:** $S_0 = 100$, $\pm \$10$ per period, $r = 0$, two periods. Fund manager: $\mathbb{P}(\text{up}) = 0.8$. Market maker: risk-neutral $\mathbb{Q}(\text{up}) = 0.5$.
- **Radon–Nikodym process** $d\mathbb{Q}/d\mathbb{P}$ [S4]:

Node	$t = 0$	$t = 1$	$t = 2$
up-up ($S = 120$)		up: $0.5/0.8 = 5/8$	$25/64$
middle ($S = 100$)	1		$(5/8)(5/2) = 25/16$
down-down ($S = 80$)		down: $0.5/0.2 = 5/2$	$25/4$

- **Expectations [S5]:**

- $\mathbb{E}^{\mathbb{P}}[S_2] = 0.8^2(120) + 2(0.8)(0.2)(100) + 0.2^2(80) = 76.8 + 32 + 3.2 = 112$.
- $\mathbb{E}^{\mathbb{Q}}[S_2] = 0.25(120) + 0.5(100) + 0.25(80) = 100$.
- **Enforceable vs not** [S5]: the market maker can lock in 100 (borrow 100, buy the stock, owe exactly 100 at $t = 2$). The manager's 112 carries risk and cannot be enforced.
- **Change-of-measure formulas** [S5–S6]:

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}\left[X \frac{d\mathbb{Q}}{d\mathbb{P}}\right], \quad \mathbb{E}^{\mathbb{P}}[X] = \mathbb{E}^{\mathbb{Q}}\left[X \frac{d\mathbb{P}}{d\mathbb{Q}}\right].$$

Check [S6]: $0.64(120)(25/64) + 0.32(100)(25/16) + 0.04(80)(25/4) = 30 + 50 + 20 = 100 \checkmark$.

- **Interpretation** [S6]: $d\mathbb{Q}/d\mathbb{P} < 1$ on high prices and > 1 on low prices: it moves weight toward the low outcomes, bringing the expectation down to the risk-neutral level.
- *[Beyond slides]* Sanity check $\mathbb{E}^{\mathbb{P}}[d\mathbb{Q}/d\mathbb{P}] = 0.64(25/64) + 0.32(25/16) + 0.04(25/4) = 0.25 + 0.5 + 0.25 = 1 \checkmark$. The process $Z_n = \mathbb{E}_n^{\mathbb{P}}[d\mathbb{Q}/d\mathbb{P}]$ (the tree values) is a $\mathbb{P}$ -martingale.
- **Traps:** inverting the ratio ($d\mathbb{Q}/d\mathbb{P} = \mathbb{Q}/\mathbb{P}$, not $\mathbb{P}/\mathbb{Q}$); computing it per step but forgetting to multiply along the path.

1.3 Arithmetic (normal) binomial tree and the CLT [S7–S9]

- **Goal** [S7] ($r = 0$): $\mathbb{E}[S_T] = S_0$ and $\mathbb{V}[S_T] = \sigma^2 T$.
- **Construction** [S7]: n steps of length T/n ; each step has mean 0 and variance $\sigma^2 T/n$, so the stock moves $\pm \sigma \sqrt{T/n}$ with probability $\frac{1}{2}$:

$$S_T = S_0 + \sum_{i=1}^n \sigma \sqrt{\frac{T}{n}} X_i, \quad X_i = \pm 1.$$

- **CLT** [S8]: i.i.d. X_i with mean μ , variance s^2 , $Y_n = \sum X_i$: $(Y_n - n\mu)/\sqrt{ns^2} \rightarrow N(0, 1)$. Here $\mu = 0$, $s^2 = 1$, so $\frac{1}{\sqrt{n}} \sum X_i \rightarrow Z$ and

$$S_T = S_0 + \sigma \sqrt{T} Z, \quad Z \sim N(0, 1).$$

- **Slide example** [S9]: $\mathbb{E}[S_T] = S_0 + \sigma \sqrt{T} \mathbb{E}[Z] = S_0$; $\mathbb{V}[S_T] = \sigma^2 T \mathbb{V}[Z] = \sigma^2 T \checkmark$.
- **Why** $\sqrt{T/n}$ *[Beyond slides]*: the step must shrink like $\sqrt{\Delta t}$ so the variance adds up to $\sigma^2 T$. Steps of Δt would give zero variance; steps of size 1 would explode. This is exactly the scaling of Brownian motion (S3).
- **Weakness** *[Beyond slides]*: S_T can be negative (same issue as S4's arithmetic model).

1.4 Non-zero interest rates and the no-arbitrage condition [S10]

- **Rate** [S10]: \$1 today grows to $\$(1+r)$ next period.
- **No-arbitrage condition** [S10]: $S_1^d < S_0(1+r) < S_1^u$ (equivalently $d < 1+r < u$).
 - *[Beyond slides]* If $S_0(1+r) \leq S_1^d$: borrow S_0 , buy the stock; you end with at least what you owe, strictly more in the up state.
 - *[Beyond slides]* If $S_0(1+r) \geq S_1^u$: short the stock, invest the proceeds.
- **Risk-neutral probability** [S10]: require $\mathbb{E}^*[S_1] = S_0(1+r)$:

$$p^* = \frac{S_0(1+r) - S_1^d}{S_1^u - S_1^d} \in (0, 1) \iff \text{no arbitrage.}$$

- **Pricing** *[Beyond slides]*: $V_0 = \frac{p^* V_1^u + q^* V_1^d}{1+r}$. Same as the hedge price $\Delta S_0 + B$ with $\Delta = \frac{V_1^u - V_1^d}{S_1^u - S_1^d}$, $B = \frac{uV_1^d - dV_1^u}{(u-d)(1+r)}$.
- **Trap:** forgetting to discount by $(1+r)$; using $\mathbb{E}^*[S_1] = S_0$ when $r \neq 0$.

1.5 Geometric (lognormal) binomial tree [S11–S12]

- **Multiplicative moves** [S11]: $S_1^u = uS_0$, $S_1^d = dS_0$ with $u = S_1(H)/S_0$, $d = S_1(T)/S_0$ (H/T = heads/tails).
- **Risk-neutral probability** [S11]:

$$p^* = \frac{(1+r) - d}{u - d}, \quad q^* = \frac{u - (1+r)}{u - d}.$$

It does not depend on S_0 , so it is the same at every node.

- **Cox–Ross–Rubinstein** [S11]: choose $d = 1/u$ (tree recombines symmetrically in log space). *[Beyond slides]* Usual calibration $u = e^{\sigma \sqrt{\Delta t}}$.

- **Slide example** [S12]: $S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, $r = 25\%$, European put $K = 5$, expiry $t = 2$.
 1. $p^* = (1.25 - 0.5)/(2 - 0.5) = 0.5$, $q^* = 0.5$.
 2. Terminal prices 16, 4, 1; put payoffs 0, 1, 4.
 3. $t = 1$: $V^u(S = 8) = (0.5 \cdot 0 + 0.5 \cdot 1)/1.25 = 0.40$; $V^d(S = 2) = (0.5 \cdot 1 + 0.5 \cdot 4)/1.25 = 2.00$.
 4. $t = 0$: $V_0 = (0.5 \cdot 0.40 + 0.5 \cdot 2.00)/1.25 = \mathbf{0.96}$.

[Beyond slides] Direct formula: $V_0 = \frac{1}{1.25^2} [0.25(0) + 0.5(1) + 0.25(4)] = 1.5/1.5625 = 0.96 \checkmark$. The call with $K = 5$ is $0.25(11)/1.5625 = 1.76$; put-call parity $C - P = S_0 - K/(1+r)^2 = 4 - 3.2 = 0.8 \checkmark$.
- **Implementation** [Beyond slides] (tested):

```
import numpy as np
def binomial_price(S0, K, u, d, r, N, payoff="put", american=False):
    """N-step recombining tree; r = per-period simple rate (1+r per step)."""
    p = ((1 + r) - d) / (u - d)
    assert 0 < p < 1, "no-arbitrage requires d < 1+r < u"
    j = np.arange(N + 1)
    ex = (lambda s: np.maximum(K - s, 0)) if payoff == "put" else (lambda s: np.maximum(s - K, 0))
    V = ex(S0 * u**j * d**(N - j)) # terminal payoffs, j = # ups
    for n in range(N - 1, -1, -1):
        V = (p * V[1:] + (1 - p) * V[:-1]) / (1 + r) # continuation value
        if american:
            V = np.maximum(V, ex(S0 * u**j[:n+1] * d**(n - j[:n+1])))
    return V[0]
print(binomial_price(4, 5, 2, 0.5, 0.25, 2)) # 0.96
print(binomial_price(4, 5, 2, 0.5, 0.25, 2, american=True)) # 1.36
```

1.6 American options [S13]

- **Rule** [S13]: exercise only if it pays more than continuing.

$$V_n^E = \frac{p^* V_{n+1}^u + q^* V_{n+1}^d}{1 + r}, \quad V_n^A = \max\left(\frac{p^* V_{n+1}^u + q^* V_{n+1}^d}{1 + r}, (K - S_n)^+\right) \text{ (put)}.$$

- **Slide example** [S13]: American put on the [S12] tree.
 - $S = 8$: continuation 0.40, exercise $(5 - 8)^+ = 0 \Rightarrow 0.40$.
 - $S = 2$: continuation 2.00, exercise $5 - 2 = 3 \Rightarrow \mathbf{3.00}$ (exercise early).
 - $t = 0$: continuation $(0.5 \cdot 0.40 + 0.5 \cdot 3.00)/1.25 = 1.36$; exercise $5 - 4 = 1 \Rightarrow V_0^A = \mathbf{1.36}$.
 - Early-exercise premium = $1.36 - 0.96 = 0.40$.
- [Beyond slides] An American call on a non-dividend stock with $r > 0$ is never exercised early (its value equals the European call); American puts can be.
- **Traps**: comparing with the exercise value only at $t = 0$ (you must check every node); using the European values at $t = 1$ when rolling back.

1.7 CLT for products: the lognormal tree [S14–S17]

- **CLT for products** [S14]: $W = X_1 X_2 \cdots X_n$ with $X_i > 0 \Rightarrow \ln W = \sum \ln X_i$, a sum of independent terms. If each has finite mean and variance, $\ln W$ is approximately normal, so W is approximately lognormal.
- **Targets** [S15] (risk-neutral, r continuous): $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$ and $\mathbb{V}[\ln S_T] = \sigma^2 T$.
- **Log tree** [S15]:

$$\ln S_i = \ln S_{i-1} + \left(r - \frac{1}{2}\sigma^2\right) \frac{T}{n} + \sigma \sqrt{\frac{T}{n}} X_i \Rightarrow \ln S_T = \ln S_0 + \left(r - \frac{1}{2}\sigma^2\right) T + \sigma \sqrt{T} \frac{1}{\sqrt{n}} \sum X_i.$$

- **Limit** [S16]: $S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right) T + \sigma \sqrt{T} Z\right]$.
- **Slide example** [S16]: $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$, $\mathbb{V}[\ln S_T] = \sigma^2 T \checkmark$ (mean and variance of an affine function of Z).
- **Slide example** [S17]: $\mathbb{E}[S_T] = S_0 e^{(r - \frac{1}{2}\sigma^2)T} \mathbb{E}[e^{\sigma \sqrt{T} Z}] = S_0 e^{(r - \frac{1}{2}\sigma^2)T} e^{\frac{1}{2}\sigma^2 T} = S_0 e^{rT}$ (MGF of Z at $\theta = \sigma \sqrt{T}$). The stock grows at the risk-free rate on average: the no-arbitrage requirement under $\mathbb{Q}$.
- **Why** $-\frac{1}{2}\sigma^2$ [Beyond slides]: it cancels the lognormal convexity $e^{\frac{1}{2}\sigma^2 T}$ (S1). Without it, $\mathbb{E}[S_T] = S_0 e^{(r + \frac{1}{2}\sigma^2)T}$, violating no-arbitrage.
- **Convergence to Black–Scholes** [Beyond slides]: CRR with $u = e^{\sigma \sqrt{\Delta t}}$, $d = 1/u$, $1 + r = e^{r \Delta t}$; call $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$: $N = 10$: 10.2534; $N = 100$: 10.4306; $N = 1000$: 10.4486; Black–Scholes 10.4506.

- **⚠ [S15]** The $\pm\sigma\sqrt{T/n}$ log tree with probability $\frac{1}{2}$ is only risk-neutral in the limit: one step gives $\mathbb{E}[S_i/S_{i-1}] = e^{(r-\frac{1}{2}\sigma^2)\Delta t} \cosh(\sigma\sqrt{\Delta t}) = e^{r\Delta t}(1 + O(\Delta t^2))$ [Beyond slides].
- **⚠** r here is continuously compounded; in [S10–S13] it is a per-period simple rate.

1.8 Random variables vs measures: the coin example [S18]

- **Setup [S18]:** toss a coin 3 times; X = number of heads, Y = number of tails.
- **No measure needed to define them [S18]:** they are functions on $\Omega = \{HHH, \dots, TTT\}$, e.g. $X(HHT) = 2$, $Y(HHT) = 1$; always $Y = 3 - X$ [Beyond slides].
- **Distributions do need a measure:**
 - Fair coin: $X, Y \sim \text{Bin}(3, \frac{1}{2})$, same distribution (probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$).
 - $p = \frac{2}{3}$ [Beyond slides]: $P(X = 0, 1, 2, 3) = \frac{1}{27}, \frac{6}{27}, \frac{12}{27}, \frac{8}{27}$; Y has these reversed. $\mathbb{E}[X] = 2$, $\mathbb{E}[Y] = 1$.
- **Expectation on a finite space [S18]:** $\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega)$.
- **Lesson:** changing the measure changes distributions and expectations, never the random variables (payoffs) themselves.

1.9 Conditional expectation [S19–S20]

- **One-step pricing as a conditional expectation [S19]:**

$$\mathbb{E}_n^*[S_{n+1}] = p^* S_{n+1}^u + q^* S_{n+1}^d, \quad S_n = \frac{1}{1+r} \mathbb{E}_n^*[S_{n+1}].$$

$\mathbb{E}_n^*$ = expectation under $\mathbb{Q}$ given the first n tosses (the information at t_n). It is itself a random variable (it depends on which node you are at).

- **Properties [S20]:**

Rule	Statement	Condition
Linearity	$\mathbb{E}_n[aX + bY] = a\mathbb{E}_n[X] + b\mathbb{E}_n[Y]$	constants a, b
Taking out what is known	$\mathbb{E}_n[XY] = X\mathbb{E}_n[Y]$	X depends only on the first n tosses
Iterated expectation (tower)	$\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X]$	$0 \leq n \leq m \leq N$
Independence	$\mathbb{E}_n[X] = \mathbb{E}[X]$	X depends only on toss $n+1$

- **Tip [Beyond slides]:** every martingale proof in S2–S4 is these four rules plus “increments are independent”.
- **Trap:** using the tower rule in the wrong order: $\mathbb{E}_m[\mathbb{E}_n[X]] = \mathbb{E}_n[X]$ too (for $n \leq m$, $\mathbb{E}_n[X]$ is already known at m), but $\mathbb{E}_n[\mathbb{E}_m[X]]$ collapses to the *earlier* time.

1.10 Martingales and the risk-neutral pricing formula [S21–S23]

- **Discounted stock is a $\mathbb{Q}$ -martingale [S21]:** divide $S_n = \frac{1}{1+r} \mathbb{E}_n^*[S_{n+1}]$ by $(1+r)^n$:

$$\frac{S_n}{(1+r)^n} = \mathbb{E}_n^*\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right], \quad M_n = \mathbb{E}_n^*[M_{n+1}], \quad M_0 = \mathbb{E}^*[M_N].$$

The last step uses the tower rule repeatedly.

- **Proof [S22]** (needs $0 < d < 1+r < u$):

$$\mathbb{E}_n^*\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] = \frac{S_n}{(1+r)^n} \cdot \frac{1}{1+r} \mathbb{E}_n^*\left[\frac{S_{n+1}}{S_n}\right] = \frac{S_n}{(1+r)^n} \cdot \frac{p^*u + q^*d}{1+r} = \frac{S_n}{(1+r)^n},$$

since $p^*u + q^*d = 1+r$ [Beyond slides] (plug in p^*). Steps: take out what is known (S_n), then independence (S_{n+1}/S_n depends only on toss $n+1$).

- **Risk-neutral pricing formula [S23]:** for a payoff V_N at t_N , the discounted derivative price is a $\mathbb{Q}$ -martingale and

$$V_n = \mathbb{E}_n^*\left[\frac{V_N}{(1+r)^{N-n}}\right].$$

- **Check on [S12] tree [Beyond slides]:** $M_0 = 4$; $\mathbb{E}^*[M_1] = (0.5 \cdot 8 + 0.5 \cdot 2)/1.25 = 4 \checkmark$.
- **Interpretation [Beyond slides]:** under $\mathbb{Q}$, every traded asset earns the risk-free rate on average, so discounting at r is correct. Under $\mathbb{P}$ the discounted stock drifts up (risk premium) and is not a martingale.
- **Traps:** claiming S_n (undiscounted) is a martingale when $r > 0$; using $\mathbb{P}$ in the pricing formula; discounting by $(1+r)^N$ instead of $(1+r)^{N-n}$ at time n .

1.11 Complete markets and arbitrage [S24–S25]

- **Complete market** [S24]: every derivative can be replicated from basic instruments (cash and the underlying), usually by dynamic rebalancing.
- **Incomplete market** [S24]: e.g. with trading frictions (or the trinomial model of S1), replication is impossible.
- **Two sides of one coin** [S24]: replicate the option from stock and cash, or hedge the option with stock to create cash.
- **Claims on** [S25]:
 - Absence of arbitrage implies the existence of risk-neutral probabilities.
 - Real world: riskier assets have higher expected returns. Risk-neutral world: all assets grow at the risk-free rate.
 - The denominator (numeraire) need not be the bank account; the choice is arbitrary.
- **The fundamental theorems** [Beyond slides]:
 1. No arbitrage $\iff$ there exists at least one equivalent martingale measure $\mathbb{Q}$.
 2. The market is complete $\iff$ that $\mathbb{Q}$ is unique.
 Binomial: unique $p^* \Rightarrow$ complete. Trinomial: a range of $p^* \Rightarrow$ incomplete, price interval (S1).
- $\triangle!$ [S25] “In a complete market ...there is no arbitrage”: completeness does not imply no-arbitrage; they are separate conditions (theorems 1 and 2 above).
- $\triangle!$ [S25] “Under the risk-neutral measure ...the relative value of any two assets is a martingale” / “all relative price processes should be a martingale”: prices divided by *the bank account* are $\mathbb{Q}$ -martingales. Dividing by a different asset (numeraire) gives martingales under a *different* measure associated with that numeraire [Beyond slides]. That is what “the choice is arbitrary” means.

2 Comparison tables

2.1 Real-world $\mathbb{P}$ vs risk-neutral $\mathbb{Q}$

	$\mathbb{P}$ (empirical)	$\mathbb{Q}$ (risk-neutral)
Up probability [S4]	0.8 (a view)	0.5 (from no-arbitrage)
Expected stock growth	risk-free + risk premium	risk-free rate
$\mathbb{E}[S_2]$ in [S5]	112 (not enforceable)	100 (enforceable by borrow-and-buy)
Discounted price	not a martingale	martingale [S21]
Used for	forecasting, risk management	pricing [S23]
Link	$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[X \, d\mathbb{Q}/d\mathbb{P}]$; requires equivalence	

2.2 Arithmetic vs geometric tree

	Arithmetic [S7–S9]	Geometric [S11, S14–S17]
Step	$S \pm \sigma \sqrt{T/n}$	$S \times u$ or $S \times d$; log step $(r - \frac{1}{2}\sigma^2)\frac{T}{n} \pm \sigma \sqrt{T/n}$
CLT applied to	sum of steps	sum of log steps
Limit	$S_T = S_0 + \sigma \sqrt{T}Z$ (normal)	$S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \sqrt{T}Z}$ (lognormal)
Mean	S_0 ($r = 0$)	$S_0 e^{rT}$
Negative prices	possible	impossible
Continuous-time analogue	$dS = \sigma \, dW$ (S4)	$dS = rS \, dt + \sigma S \, dW$ (GBM)

2.3 European vs American on a tree

	European	American
Node value	$\frac{p^*V^u + q^*V^d}{1+r}$	$\max(\text{continuation, exercise})$
Closed form	$\mathbb{E}^*[V_N]/(1+r)^N$	none; backward induction only
[S12–S13] put	0.96	1.36 (exercise at $S = 2$)
Relation	$V^A \geq V^E$	premium = value of the early-exercise right

2.4 Conditional expectation rules [S20]

Linearity	$\mathbb{E}_n[aX + bY] = a\mathbb{E}_n[X] + b\mathbb{E}_n[Y]$
Known factor	$\mathbb{E}_n[XY] = X\mathbb{E}_n[Y]$ if X is known at n
Tower	$\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X]$, $n \leq m$
Independence	$\mathbb{E}_n[X] = \mathbb{E}[X]$ if X is independent of the first n tosses

3 Master formula sheet

Change of measure [S2–S6]

- $\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)}$; equivalence $\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0$.
- $\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[X \frac{d\mathbb{Q}}{d\mathbb{P}}]$; $\mathbb{E}^{\mathbb{P}}[\frac{d\mathbb{Q}}{d\mathbb{P}}] = 1$ [Beyond slides].

Trees [S7–S13]

- Arithmetic: step $\pm\sigma\sqrt{T/n}$; $S_T \rightarrow S_0 + \sigma\sqrt{T}Z$.
- No arbitrage: $d < 1 + r < u$. $p^* = \frac{(1+r)-d}{u-d}$, $q^* = \frac{u-(1+r)}{u-d}$. CRR: $d = 1/u$ ($u = e^{\sigma\sqrt{\Delta t}}$ [Beyond slides]).
- European: $V_n = \frac{p^*V_{n+1}^u + q^*V_{n+1}^d}{1+r}$; $V_0 = \frac{1}{(1+r)^N} \sum_j \binom{N}{j} p^{*j} q^{*N-j} V_N(j)$ [Beyond slides].
- American put: $V_n = \max\left(\frac{p^*V_{n+1}^u + q^*V_{n+1}^d}{1+r}, (K - S_n)^+\right)$.
- Hedge [Beyond slides]: $\Delta_n = \frac{V_n^u - V_n^d}{S_n^u - S_n^d}$.

Lognormal limit [S14–S17]

- $S_T = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z]$; $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$; $\mathbb{V}[\ln S_T] = \sigma^2 T$; $\mathbb{E}[S_T] = S_0 e^{rT}$.

Martingales and pricing [S19–S23]

- $S_n = \frac{1}{1+r} \mathbb{E}_n^*[S_{n+1}]$; $p^*u + q^*d = 1 + r$.
- $M_n = \frac{S_n}{(1+r)^n} = \mathbb{E}_n^*[M_{n+1}]$; $M_0 = \mathbb{E}^*[M_N]$.
- $V_n = \mathbb{E}_n^*\left[\frac{V_N}{(1+r)^{N-n}}\right]$.

4 Glossary

- **American option:** exercisable at any time up to expiry [S13].
- **Backward induction:** rolling values from expiry to today node by node.
- **Central limit theorem:** standardised sums of i.i.d. variables tend to $N(0, 1)$ [S8].
- **Complete market:** every payoff can be replicated [S24].
- **Conditional expectation $\mathbb{E}_n$:** expectation given the first n tosses [S19].
- **Cox–Ross–Rubinstein (CRR):** geometric tree with $d = 1/u$ [S11].
- **Early-exercise premium:** $V^A - V^E$.
- **Equivalent measures:** agree on which events have zero probability [S3].
- **Martingale:** $M_n = \mathbb{E}_n[M_{n+1}]$; best forecast of tomorrow is today [S21].
- **Numeraire:** the asset prices are divided by (bank account by default) [S25].
- **Probability space:** sample space plus probability measure [S2].
- **Radon–Nikodym derivative:** ratio $d\mathbb{Q}/d\mathbb{P}$ reweighting outcomes [S2].
- **Risk-neutral measure $\mathbb{Q}$:** measure under which discounted prices are martingales [S21–S23].
- **Risk-neutral pricing formula:** $V_n = \mathbb{E}_n^*[V_N/(1+r)^{N-n}]$ [S23].

- **Sample space** Ω : set of all outcomes (paths) [S2].
- **Tower property**: $\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X]$ for $n \leq m$ [S20].

5 Self-test

Questions

- Q1.** In the [S4] example, compute $d\mathbb{Q}/d\mathbb{P}$ on each two-step path and verify that $\mathbb{E}^{\mathbb{P}}[S_2 d\mathbb{Q}/d\mathbb{P}] = 100$ and $\mathbb{E}^{\mathbb{P}}[d\mathbb{Q}/d\mathbb{P}] = 1$.
- Q2 (conceptual).** What does it mean for $\mathbb{P}$ and $\mathbb{Q}$ to be equivalent, and why is it needed?
- Q3.** In an arithmetic tree with n steps and $r = 0$, why is each step $\pm\sigma\sqrt{T/n}$? Show $S_T = S_0 + \sigma\sqrt{T}Z$ has the required mean and variance.
- Q4.** $S_0 = 50$, $u = 1.2$, $d = 0.9$, $r = 5\%$ per period. Check no-arbitrage, find p^* , and price a one-period call with $K = 52$. Give the hedge.
- Q5 ([S12]).** $S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, $r = 25\%$, $K = 5$, two periods. Price the European put and call; check put–call parity.
- Q6 ([S13]).** Price the American put on the same tree. Where is it optimal to exercise?
- Q7 ([S16–S17]).** For $S_T = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z]$, find $\mathbb{E}[\ln S_T]$, $\mathbb{V}[\ln S_T]$ and $\mathbb{E}[S_T]$. Why the $-\frac{1}{2}\sigma^2$?
- Q8 ([S18]).** Three tosses, $X = \text{heads}$, $Y = \text{tails}$. Give $X(\omega), Y(\omega)$ for all outcomes, and the distributions under $p = \frac{1}{2}$ and $p = \frac{2}{3}$.
- Q9 ([S22]).** Prove that $S_n/(1+r)^n$ is a $\mathbb{Q}$ -martingale, naming each conditional-expectation rule you use.
- Q10 (conceptual).** State the two fundamental theorems of asset pricing and use them to classify the binomial and trinomial models. What is wrong with “complete market $\Rightarrow$ no arbitrage”?

Solutions

- S1.** Paths: UU 25/64, UD and DU 25/16, DD 25/4. $0.64(120)(25/64) + 0.32(100)(25/16) + 0.04(80)(25/4) = 30 + 50 + 20 = 100$. $0.64(25/64) + 0.32(25/16) + 0.04(25/4) = 0.25 + 0.5 + 0.25 = 1$.
- S2.** They agree on which outcomes are possible: $\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0$. Otherwise the ratio is undefined somewhere and information about one measure is lost; economically, an arbitrage under one measure would not be one under the other.
- S3.** Each step needs variance $\sigma^2 T/n$ so that n independent steps sum to $\sigma^2 T$; a $\pm a$ step with probability $\frac{1}{2}$ has variance a^2 , so $a = \sigma\sqrt{T/n}$. $\mathbb{E}[S_T] = S_0$, $\mathbb{V}[S_T] = \sigma^2 T \mathbb{V}[Z] = \sigma^2 T$.
- S4.** $0.9 < 1.05 < 1.2 \checkmark$. $p^* = (1.05 - 0.9)/0.3 = 0.5$. Payoffs: $60 - 52 = 8$, $(45 - 52)^+ = 0$. $C_0 = 0.5 \cdot 8/1.05 = 3.81$. Hedge: $\Delta = 8/15 = 0.533$ shares, bond $= C_0 - \Delta S_0 = 3.81 - 26.67 = -22.86$ (borrow 22.86).
- S5.** $p^* = 0.5$. Put: payoffs 0, 1, 4 $\Rightarrow$ 0.40, 2.00 $\Rightarrow$ 0.96. Call: payoffs 11, 0, 0 $\Rightarrow$ $0.25 \cdot 11/1.5625 = 1.76$. Parity: $1.76 - 0.96 = 0.80 = 4 - 5/1.5625 \checkmark$.
- S6.** Node $S = 2$: $\max(2.00, 3) = 3$ (exercise). Node $S = 8$: $\max(0.40, 0) = 0.40$. Today: $\max((0.2 + 1.5)/1.25, 1) = \max(1.36, 1) = 1.36$. Exercise at $t = 1$ after a down move.
- S7.** $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$; $\mathbb{V} = \sigma^2 T$; $\mathbb{E}[S_T] = S_0 e^{(r - \frac{1}{2}\sigma^2)T} e^{\frac{1}{2}\sigma^2 T} = S_0 e^{rT}$. The $-\frac{1}{2}\sigma^2$ offsets the lognormal convexity so the stock grows at r under $\mathbb{Q}$.
- S8.** HHH (3,0); HHT, HTH, THH (2,1); HTT, THT, TTH (1,2); TTT (0,3). Fair: both $\text{Bin}(3, \frac{1}{2})$: $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$. $p = \frac{2}{3}$: X : $\frac{1}{27}, \frac{6}{27}, \frac{12}{27}, \frac{8}{27}$ for 0..3; Y reversed.
- S9.** $\mathbb{E}_n^*[S_{n+1}/(1+r)^{n+1}] = \frac{S_n}{(1+r)^{n+1}} \mathbb{E}_n^*[S_{n+1}/S_n]$ (known factor S_n) $= \frac{S_n}{(1+r)^{n+1}} \mathbb{E}^*[S_{n+1}/S_n]$ (independence: depends only on toss $n+1$) $= \frac{S_n}{(1+r)^{n+1}} (p^*u + q^*d) = \frac{S_n}{(1+r)^n}$ since $p^*u + q^*d = 1 + r$.
- S10.** (1) No arbitrage $\iff$ an equivalent martingale measure exists. (2) Complete $\iff$ it is unique. Binomial with $d < 1 + r < u$: unique $p^* \Rightarrow$ arbitrage-free and complete. Trinomial: many $p^* \Rightarrow$ arbitrage-free but incomplete. A complete market can still contain an arbitrage: take a binomial tree with $u < 1 + r$. Every payoff is still replicable (two states, two assets), but shorting the stock and lending the proceeds wins in every state, and no $p^* \in (0, 1)$ exists.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Probability space; changing measure; Radon–Nikodym derivative	1.1
S3	When the derivative exists; equivalence of measures	1.1
S4	Example: fund manager (0.8) vs market maker (0.5); $d\mathbb{Q}/d\mathbb{P}$ tree	1.2
S5	$\mathbb{E}^{\mathbb{P}}[S_2] = 112$, $\mathbb{E}^{\mathbb{Q}}[S_2] = 100$; change-of-measure formulas	1.2
S6	Verification of both formulas	1.2
S7	Arithmetic binomial tree construction	1.3
S8	Central limit theorem; normal limit	1.3
S9	Example: mean and variance of $S_0 + \sigma\sqrt{T}Z$	1.3
S10	Non-zero rates; $S^d < S_0(1+r) < S^u$; p^*	1.4
S11	Geometric tree; u, d ; p^* ; CRR $d = 1/u$	1.5
S12	Example: European put on a 2-step tree	1.5
S13	American options; example	1.6
S14	CLT for products $\Rightarrow$ lognormal	1.7
S15	Log-price tree with drift $r - \frac{1}{2}\sigma^2$	1.7
S16	Lognormal limit; example on $\mathbb{E}[\ln S_T]$, $\mathbb{V}[\ln S_T]$	1.7
S17	Example: $\mathbb{E}[S_T] = S_0 e^{rT}$ via MGF	1.7
S18	Coin example; random variables vs measure; finite expectation	1.8
S19	Conditional expectation; $S_n = \mathbb{E}_n^*[S_{n+1}]/(1+r)$	1.9
S20	Properties of conditional expectation	1.9
S21	Discounted stock is a martingale; $M_0 = \mathbb{E}^*[M_N]$	1.10
S22	Proof of the martingale property	1.10
S23	Risk-neutral pricing formula	1.10
S24	Complete vs incomplete markets; replication	1.11
S25	No arbitrage $\Rightarrow \mathbb{Q}$; risk-free growth; numeraires	1.11

Appendix B. Flag summary ($\triangle!$)

- [S2] Probability space is $(\Omega, \mathcal{F}, \mathbb{P})$; $\mathcal{F}$ omitted because Ω is finite.
- [S3] With $\mathbb{P}(\omega) = 0 < \mathbb{Q}(\omega)$ it is $d\mathbb{Q}/d\mathbb{P}$ that is undefined, not $\mathbb{P}/\mathbb{Q}$ (which is 0).
- [S10 vs S15–S17] r is a per-period simple rate in the tree slides but continuously compounded in the lognormal-limit slides.
- [S11] $u = S_1(H)/S_0$ uses coin notation (H/T) introduced only on [S18].
- [S15] The $\pm\sigma\sqrt{T/n}$ log tree is risk-neutral only in the limit $n \rightarrow \infty$.
- [S25] “Complete market $\Rightarrow$ no arbitrage” is not a theorem; no-arbitrage $\iff$ existence of $\mathbb{Q}$, completeness $\iff$ uniqueness.
- [S25] Relative prices are $\mathbb{Q}$ -martingales only with the bank account as denominator; another numeraire comes with its own measure.
- [S9, S12, S13, S16, S17] Examples are posed without worked answers; the solutions here are my own.